

What is KATO?

KATO is a simple, Python-inspired programming language designed for beginners and educational purposes. It allows you to write code in English-like syntax with indentation-based blocks (no end statements), and it can be transpiled to Python to run in browsers (via Pyodide) or locally.

Key points:

- Easy to read and write** - uses plain English keywords (if, while, for, def, print).

- Dynamic typing** - no need to declare variable types.

- Indentation-based** - blocks are defined by indentation, like Python.

- Browser-friendly** - can run directly in the browser using a sandbox.

- Educational focus** - ideal for learning programming concepts without complex syntax.

- It's basically a friendly layer** over Python for experimenting and teaching coding.

KATO User Manual

KATO Quick Manual (Browser / Pyodide Version)

1. Variables and Assignments:

```
x = 10          # number
name = "Ivan"   # text
isStudent = True # boolean
```

*No type declaration is needed.

*Lists are defined just like in Python:

```
numbers = [1, 2, 3, 4, 5]
```

2. Operators:

```
# Arithmetic
sum = 5 + 3
diff = 10 - 4
mul = 6 * 7
div = 20 / 4
mod = 10 % 3
```

```
# Comparison
x == 5
x != 10
x > 0
x < 100
x >= 10
x <= 20
```

```
# Logical
a and b
a or b
not a
```

3. Print / Output:

```
print "Hello, world!"
print "The number is", x
```

*Automatically transpiled to `print("...")`.

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4. Control Structures:

```
If / Else
if x > 5:
    print "x is greater than 5"
else:
    print "x is 5 or less"
```

```
While Loop
i = 0
while i < 5:
    print i
    i = i + 1
For Loop (range style)
for i in range(0, 5):
    print i
```

5. Functions:

```
def greet(name):
    print "Hello " + name

greet("Ivan")
```

def requires : and indentation for the block.

*No end is used.

6. Lists & Iteration:

```
numbers = [1, 2, 3, 4, 5]

for n in numbers:
    print n
```

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7. User Input:

```
name = input("Enter your name: ")  
print "Hello " + name
```

*Works just like Python's input().

8. Tips for the Sandbox:

-Indent with 4 spaces (or press Tab; the editor inserts spaces).

-No end or fin: everything is controlled by indentation.

-Strings must use quotes: "text".

-Avoid Python-specific type declarations; just assign variables.

-Print statements: use print "... " in KATO → automatically transpiled.

Conclusion:

KATO is a beginner-friendly, Python-inspired language that lets you explore programming in a clear and intuitive way. With its English-like syntax and indentation-based structure, you can focus on learning concepts without worrying about complex rules. Let your imagination run wild in the sandbox, experiment with ideas, and bring your coding projects to life!